Evaluation Report

Community-School Partnership (CSP) Mentoring Program
Washougal High School | September 2024 – May 2025

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Executive Summary

This report evaluates the impact of the Community-School Partnership (CSP) mentoring program on student performance and attendance at Washougal High School. Our findings show that consistent, early mentoring leads to steady long-term academic gains—especially in core subjects where improvement may take more time. Students who received mentoring every two weeks often ended up closing the performance gap with the class average, with many showing sustained growth beyond the classes they were mentored in.

Introduction

The Community-School Partnership (CSP) is a mentoring program developed by the East County Citizens Alliance (ECCA) in collaboration with Washougal High School. Launched in 2022, the program aims to support academically struggling students through regular, course-specific mentoring provided by trained community volunteers. Students were referred by teachers or counselors if their grades in a course fell between 1.6 and 2.0 on a 4.0 scale. Mentoring sessions were offered weekly and delivered either one-on-one or in small groups, depending on availability.

This report specifically evaluates the program's impact during the 2024–2025 academic year, focusing on whether students showed measurable improvements in grades and attendance. Importantly, it examines outcomes in both classes where students received direct mentoring and those where they did not, helping assess not only the targeted benefits of the program but also any broader academic spillover. To do this, we analyzed weekly grade and attendance data, applying statistical modeling to understand patterns over time.

Some terms that appear throughout the paper include *predictors*—the variables that help explain or influence an outcome—and *parametric coefficients*, which show the estimated size and direction of each predictor's effect in the model. The *response variables*—the main outcomes we're trying to understand—are students' academic performance and attendance. These technical terms are defined where relevant in the sections that follow.

3 Data

3.1 Software and Tools

All data was cleaned in Python using pandas (Pandas Development Team, 2024). The Data Exploration and Modeling was done in the R language using the packages ggplot2 (Wickham, 2025) and dplyr (Wickham et al., 2023). The Data Exploration section also used gridExtra (Baptiste, 2017) and the grid package (part of base R). The Modeling also utilized the packages: glme (Weerahandi et al., 2021), itsadug (van Rij et al., 2022), mgcv (Wood, 2025), and mgcViz (Fasiolo et al., 2025).

3.2 Data Description and Preprocessing

We began our analysis by integrating two primary sources provided by Washougal High School and the CSP team:

- Student Records Dataset: This included weekly academic performance (GPA-based percentage and letter grades) for each student across all their enrolled courses.
- Mentoring Attendance Logs: These detailed weekly mentoring session participation, including session dates, subjects, mentors, and attendance status.

Each row in the final dataset corresponded to a unique (Student × Course × Week) combination. Tables 1 and 2 below contain descriptions of key variables used in the analysis.

Table 1
Response Variables

Column Name	Description
Grade Perc	A continuous variable representing the student's weekly grade percentage in a given course.
Grade Z Score	A standardized version of the Grade Perc that reflects how the student performed relative to the class average in that week.
Absence Z Score	A standardized version of total absences, used in attendance analysis.

Table 2
Predictor Variables

Column Name	Description
Attended	Whether the student attended mentoring that week in that course (Yes, No, or Not Applicable).
Total Sessions Attended This Week for this Class	How many mentoring sessions the student attended that week for that course (integer).
Week	Week number of the academic year (ranging from 2 to 34).
Period	Class period (1 through 6). Used as a fixed effect in modeling to account for scheduling structure.
Semester	Semester identifier (S1 or S2).
Grade Year	Student's current grade level (e.g., 9, 10, 11, 12).
Total Days Absent	Total number of days the student was absent during that week.
Mentor Course	Whether the course was ever used for mentoring for any student (Yes or No).
Mentored Student	Whether the student received mentoring in any course (Yes or No).
Running Total (Any)	Cumulative total of mentoring sessions attended throughout the year by the student across all classes.
Running Total (Class)	Cumulative total of mentoring sessions attended by the student in the specific course.
Running Total (Any, Semester Scoped)	Cumulative total of mentoring sessions attended throughout the semester by the student across all classes.
Subject	General subject category of the course (e.g., Math, English, Science). Derived from Course Code.
Course Code	Unique identifier for the course, often encoding department and section (e.g., MAT522 / 12).

Session_Date	Calendar date of the session, aligned to the Monday of the academic week.
Mentor	Name of the mentor assigned to the student (not always filled).

The preparation process, detailed in Appendix A, involved multiple stages:

1. Column Standardization and Cleaning: All column names were stripped of whitespace and unified across data sources (e.g., “Student #” and “PSU#” were both renamed to “Student ID”).
2. Combining Semester Datasets: Weekly grade pull files from both semesters were concatenated into one unified dataset. Attendance logs were cleaned and transformed from a wide format (multiple date columns) to a long format (one row per session).
3. Parsing and Aligning Dates: Session dates were parsed using regex and normalized to the corresponding Monday of each week to align with grade pull timestamps.
4. Encoding Attendance: Attendance values were cleaned and encoded as “Yes,” “No,” or “Not Applicable.” Duplicated rows were deduplicated, prioritizing “Yes” sessions when present.
5. Merging Attendance with Grades: After reshaping both datasets, we merged them by (Student ID, Course Code, Session Date) to align mentoring sessions with weekly grades.
6. Flagging Mentoring Participation: We added indicator variables for whether a course was ever mentored and whether a student ever received mentoring, based on participation history.
7. Filling Missing Weeks: For consistency, missing (Student × Course × Week) combinations were padded using previous or next week’s data as a template. Non-attendance weeks were explicitly labeled “No.”
8. Filtering Data Entry Errors: Rows with incomplete student records were removed. For example, known data-entry issues were filtered out based on staff names and expected weekly counts.
9. Subject Code Mapping: Course codes were used to infer subject areas (e.g., “MAT” → Math, “ENG” → English), enabling subject-level comparisons in the analysis.
10. Standardizing Grades and Attendance: Weekly Z-scores were computed by subtracting the weekly class mean and dividing by the class standard deviation. This provided a normalized measure of performance and attendance.

This preprocessing pipeline ensured that our final dataset was suitable for weekly longitudinal analysis and comparison across mentoring status, subjects, and grade levels.

3.3 Data Exploration

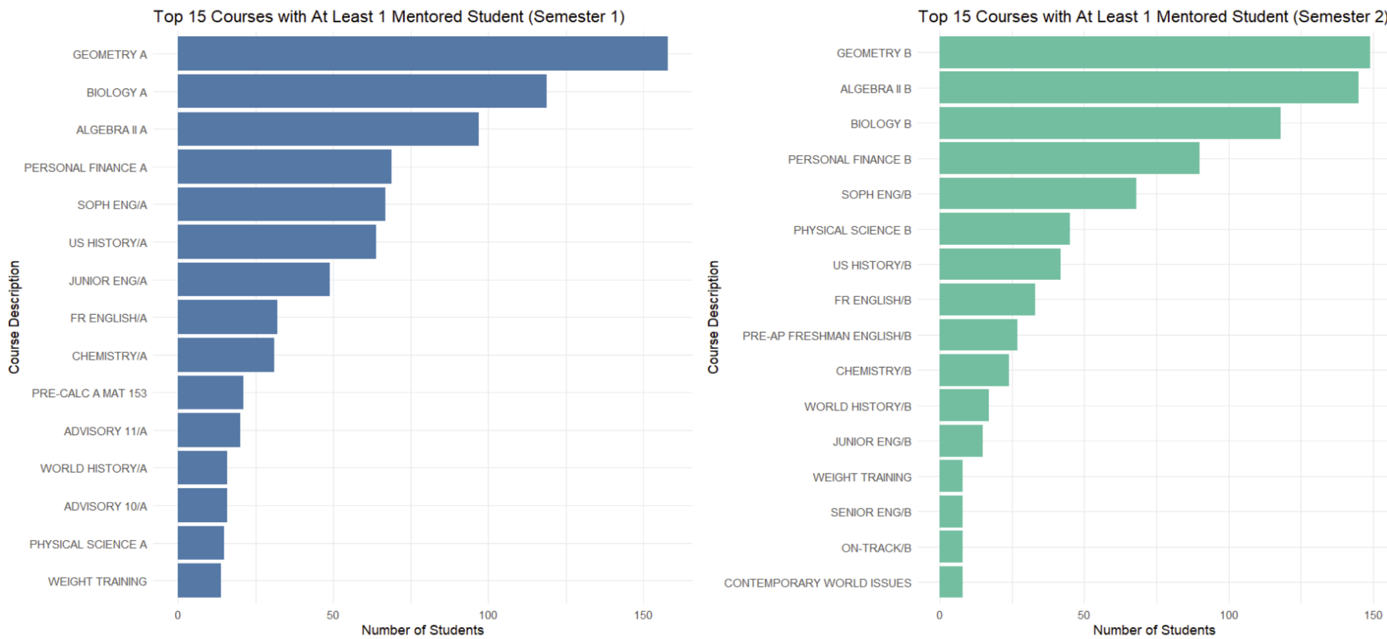

Figure 1

Top 15 Courses with At Least 1 Mentored Student for Semesters 1 & 2

One of the first patterns we observed was a strong overlap between the school's most commonly taken classes and the subjects with the most students being mentored. As shown in Figure 1, nearly all of the 15 most-enrolled classes fall within the top five most frequently mentored subjects in each semester (see Figure 2).

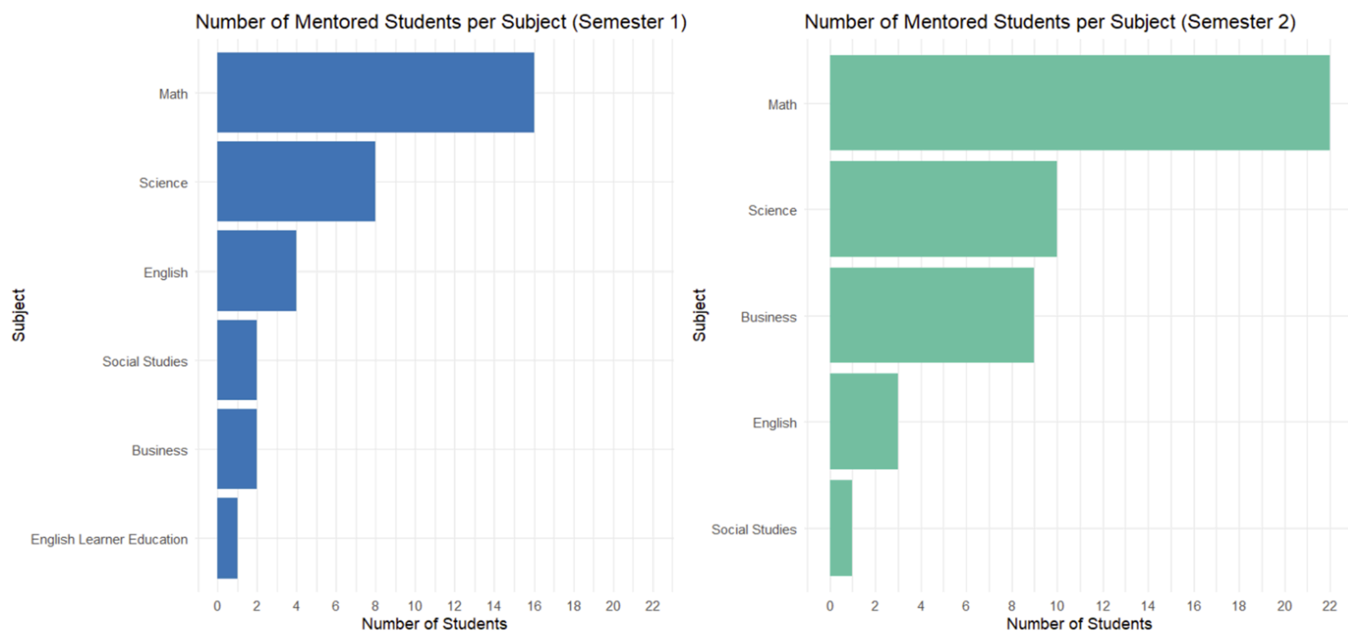

Figure 2

Number of Mentored Students per Subject for Semesters 1 & 2

Math consistently held the top spot across both semesters, followed by Science and English. In Semester 2, Business rose into the top three surpassing English, with the number of mentored students increasing from just 2 in Semester 1 to 9 in Semester 2. A closer look at the Business category (Figure 3) reveals that all of this increase came from a single course—Personal Finance—suggesting that students may have struggled more than expected in Semester 1, prompting a ramp-up in mentoring support in Semester 2. This pattern indicates that encouraging Personal Finance students to attend mentoring earlier, such as in Semester 1, could be beneficial. Geometry also stood out as the most mentored course overall, maintaining the highest number of mentored students across both semesters. We also observed that many of the most frequently mentored classes were among the school's most widely enrolled courses. For instance, Geometry, Algebra, and Biology stood out as both popular and commonly mentored. This overlap suggests that the program is focusing on subjects with high student enrollment, which may also correspond to areas of greater academic support needs.

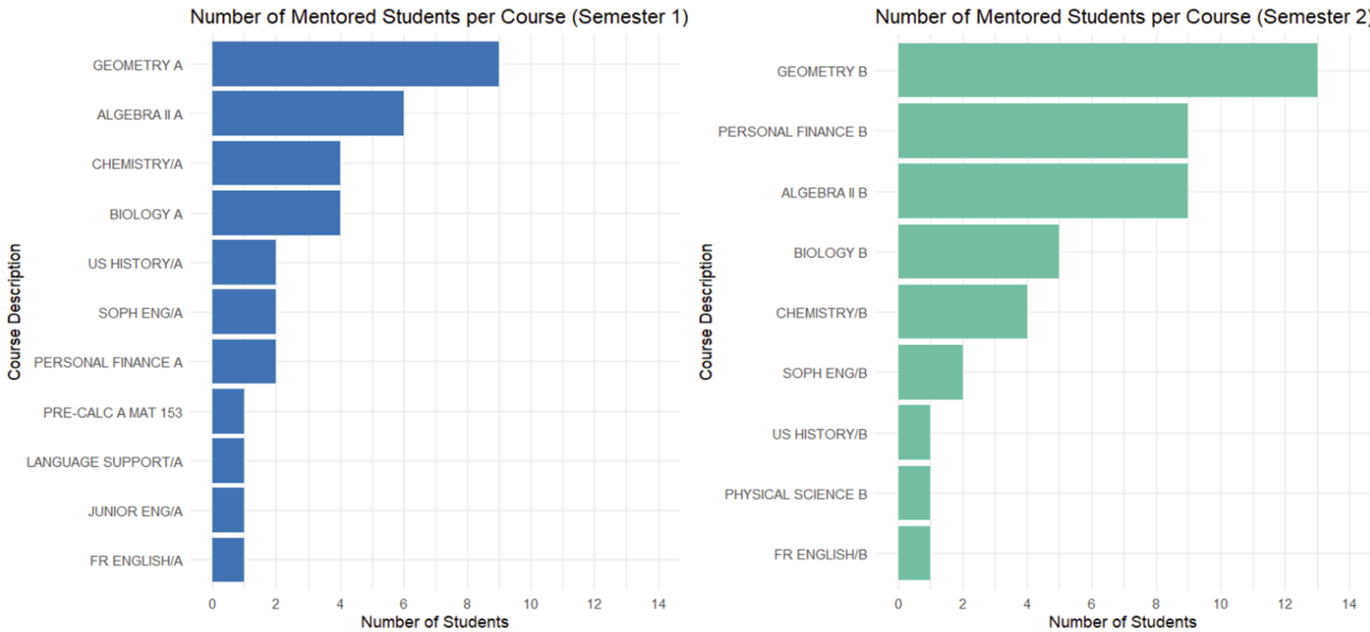

Figure 3

Number of Mentored Students per Course for Semesters 1 & 2

Looking at how often students attended mentoring sessions, most participated between 2 and 10 times per semester (as seen in Appendix B Figure B1). However, a small number attended over 20 sessions, highlighting variation in commitment levels. This may reflect differences in when students were referred, the class difficulty, or individual motivation. For details on the classes with the highest mentoring attendance, refer to Appendix B, Figure B2.

Cumulative attendance patterns over time revealed further variation in engagement with the program. Some students attended regularly each week, while others had more sporadic participation or joined late in the semester. These trends are illustrated in Figures B3 and B4 in Appendix B. These differences became more pronounced when analyzed by grade level: mentoring was more common among 10th and 11th graders, with some participation from 12th graders, while only one 9th grader received mentoring—and only during the second semester (see Appendix B, Figure B5). This pattern is further supported by a weekly attendance plot in Figure B6, which shows that sophomores and juniors maintained steady participation, whereas other grades exhibited more fluctuation, indicating inconsistent engagement within certain student groups.

4 Methods

Even though the methods presented throughout this section were implemented with both grades and attendance, in the main body of the report we present the analysis using only grades for simplicity. Raw grades and attendance can be hard to interpret on their own because classes vary in difficulty, assignments are given and weighted differently, and grades accumulate at different rates. Similarly, attendance patterns can differ based on when classes meet, how often they meet, and how absences are recorded. As a result, both grades and attendance tend to “settle” or take shape at different times depending on the course. Therefore, to properly assess the effects of mentoring on student performance over time, we converted students' grades (GPA values) and attendance (tardies) into standardized Z-scores. This approach addresses the various differences in classroom grading standards, teaching styles, and class difficulty, evening the playing field so meaningful comparisons within mentored students and effects of mentored courses could be made. This allows us to focus on the scope of how students are performing in relation to their peers each week of each class and comparing the individual trajectory from start of intervention to end. Each class mean and standard deviation was calculated on a week to week basis as grades were rolling.

$$z_{itc} = \frac{X_{itc} - \mu_{tc}}{\sigma_{tc}} \tag{1}$$

Equation 1 above shows how the z-score is calculated. z_{itc} represents the standardized score for student *i*, at week *t*, for class *c*. Additionally *X* represents the grade a mentoring student *i* received at week *t*, for a given class *c*. The class average is then subtracted from this value and we divide this difference by the class standard deviation.

A Z-score of 0 indicates a student is performing right at the class average, while positive and negative values reflect above and below-average performance within their class. The strength in using a standardized approach is that interpretation of Z-scores is very intuitive; a quick glance at a Z-score gives context to how a student is doing without the need to interpret raw grades that may be inflated or deviate wildly more than one would expect. Through a standardized lens mentors and counselors can enhance decision making with informed data on struggling students. These Z-Scores can be used proactively to flag students who need extra support if they’re struggling throughout the term. Conversely an upwards or steadying effect helps validate the effectiveness of mentoring’s effect on students.

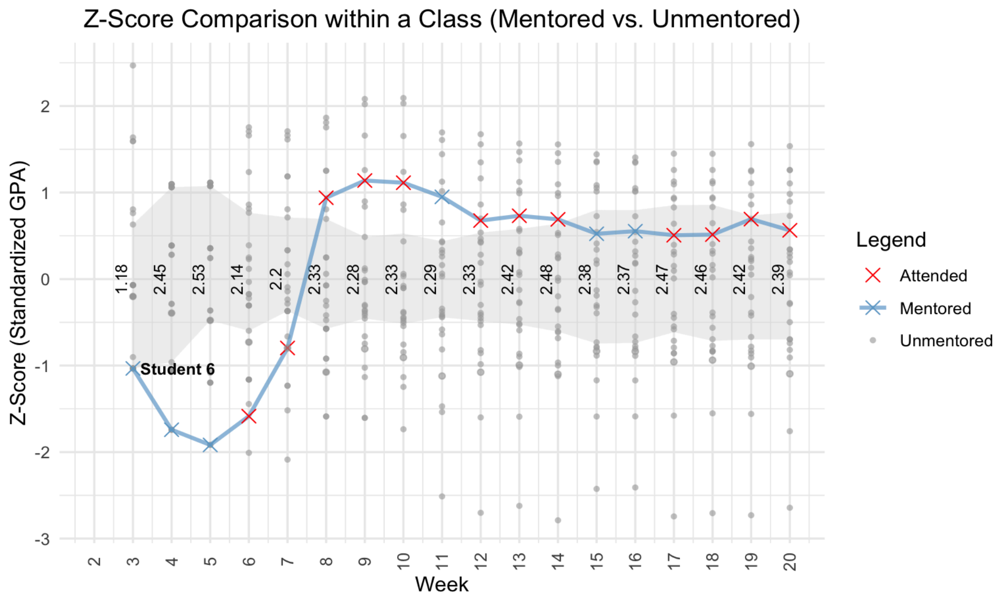

Figure 4

Grade Z-Score Comparison within a Class (Mentored vs. Unmentored)

Figure 4 compares a mentored student's performance within a class in this case Geometry A taught by Rochelle Aiton in period 2. The grey ribbon represents the middle 50% of the classroom's performance each week, essentially making it a visual peer comparison tool. So when a student is above this area they are performing better than most of their classmates. Student 6 has their trajectory connected in blue to contrast their performance among their unmentored peers. The figure highlights individual variability, group trends, and patterns in mentoring session attendance. At the start they are performing worse than the class average. However, after the first couple mentoring sessions, we see an upwards trend that eventually stabilizes either above or near the mean indicating that consistent mentoring may play a role in improving and maintaining a student's performance within a class.

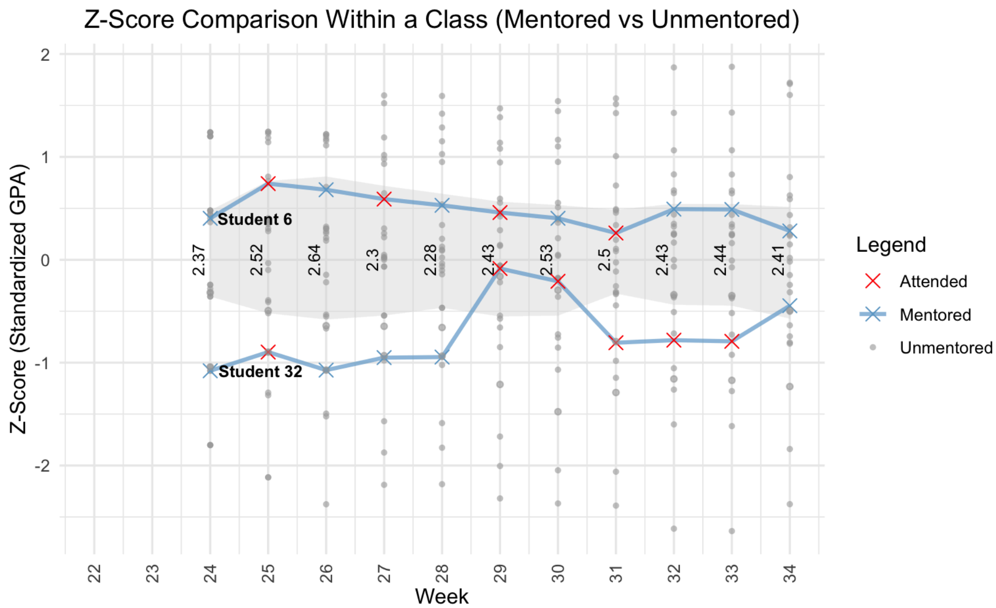

Figure 5

Grade Z-Score Comparison Within a Class (Mentored vs Unmentored)

Figure 5 displays Z-score trajectories for Semester 2, for Geometry B the course after Geometry A. Also taught by Rochelle Aiton. We follow student 6 continuing the comparison between mentored and non-mentored students. Student 6 maintains performance levels consistently above the class average, reflecting ongoing progress following earlier mentoring support in semester 1. In addition, Student 32's performance is also tracked in this plot. It's interesting to note that Student 32 was also in Geometry A which was taught by Rochelle the semester prior with the one difference from Student 6–it was for Period 4.

Student 6 began Geometry A with an F**,** but after receiving mentoring, showed significant improvement. In Geometry B, their performance remains consistently above the class average, indicating steady academic growth and the lasting impact of mentoring support from the previous semester.

Table 3

Geometry A Performance of Students 6 and 32 at Weeks 4 and 20

Student.ID	Course.Code	Course.Description	Week	Grade Perc	Staff Name
6	MAT522 / 12	Geometry A	4	0.00	Rochelle Aiton
32	MAT522 / 14	Geometry A	4	1.90	Rochelle Aiton
6	MAT522 / 12	Geometry A	20	2.90	Rochelle Aiton
32	MAT522 / 14	Geometry A	20	2.26	Rochelle Aiton

In contrast, as seen in Table 3, Student 32, who also took Geometry A with Rochelle (in Period 4 and without mentoring in the first semester), started with a D. Despite having a slightly stronger initial grade than Student 6, their Z-score trajectory in Geometry B ended up being lower. This contrast suggests that mentoring in Semester 1 may have played a key role in preparing Student 6 for sustained success, while the absence of support left Student 32 with less academic momentum. When viewing the plot for semester 2, we also see that Student 6 continued to do better relative to their peers compared to Student 32.

To better understand these individual patterns and group-level effects, we applied Generalized Additive Models (GAMs). Instead of assuming a straight-linear relationship (like in simple models), GAMs let each variable's influence be represented as a curve. This helps us capture patterns in data without needing to know the exact shape of the relationship between the response and the variable ahead of time. These models are particularly useful in helping understand different factors that affect student outcomes even when the patterns are non-linear. This makes it flexible in incorporating fixed effects as well as capturing more realistic trends like gradual improvement, plateaus or sudden changes. Specifically, instead of assuming each variable has a fixed, linear impact, GAMs let us model smooth curves over time effectively tracking how weekly attendance or mentoring sessions influence Z-score trajectories. We referenced materials from QCBS (2023) to assist with the modeling.

In our model some of the columns we used as fixed effects are Mentor.Course, Grade.Year, Subject(Course.Category), Period and Semester to account for the structured influences across students. Other variables such as Week, were included with a smooth. Meaning instead of forcing linear patterns on something that is likely more complicated and nuanced, we can flexibly allow it to curve. This is important because students do not always progress in a fully linear path towards progression. We also included a smooth term for Running.Total..Class., which tracks the number of mentoring sessions a student has received in a class. This allows us to explore whether consistent mentoring leads to a steady improvement or if effects plateau off after a certain threshold.

5 Results

```
Z.Score ~ Mentor.Course * Grade.Year + Mentor.Course * Course.Category +
    s(Week.sem, k = 15) + Period + Semester + s(Running.Total..Class.,
    k = 6) + s(Total.Days.Absent, k = 6) + s(Student.ID, Week.sem,
    bs = "re") + s(Student.ID, bs = "re")
```

A. parametric coefficients	Estimate	Std. Error	t-value	p-value
(Intercept)	-0.5079	0.1156	-4.3923	< 0.0001
Mentor.CourseYes	2.9713	0.4036	7.3621	< 0.0001
Grade.Year	0.0442	0.0105	4.2041	< 0.0001
Course.CategoryMath	-0.3849	0.0305	-12.6122	< 0.0001
Course.CategoryScience	-0.1185	0.0256	-4.6243	< 0.0001
Course.CategoryEnglish	-0.0860	0.0206	-4.1800	< 0.0001
Course.CategoryBusiness	-0.4116	0.0512	-8.0388	< 0.0001
Period	-0.0088	0.0037	-2.3743	0.0176
SemesterS2	0.0507	0.0158	3.2036	0.0014
Mentor.CourseYes:Grade.Year	-0.3034	0.0358	-8.4667	< 0.0001
Mentor.CourseYes:Course.CategoryMath	-0.0774	0.0956	-0.8099	0.4180
Mentor.CourseYes:Course.CategoryScience	-0.0115	0.0975	-0.1179	0.9061
Mentor.CourseYes:Course.CategoryEnglish	-0.7027	0.1182	-5.9468	< 0.0001
Mentor.CourseYes:Course.CategoryBusiness	0.3239	0.1147	2.8230	0.0048
B. smooth terms	edf	Ref.df	F-value	p-value
s(Week.sem)	3.5305	4.4058	16.0587	< 0.0001
s(Running.Total..Class.)	3.2469	3.8143	7.0151	< 0.0001
s(Total.Days.Absent)	4.2651	4.7026	183.7784	< 0.0001
s(Student.ID,Week.sem)	0.0006	1.0000	0.0001	0.7658
s(Student.ID)	0.1080	1.0000	0.1108	0.3109

Figure 6

Modeling Equation and Parametric Coefficients Table Summary

In our model for attendance and grades, we combine linear effects such as mentoring status, grade year, course subject, period, and other relevant factors. All the highlighted coefficients are statistically significant at an $\alpha = 0.05$, indicating that those predictors have a strong influence on the response variable. The coefficients highlighted in yellow are more straightforward to interpret, while the ones in blue are more complex because they involve either interaction effects—where the impact of one variable depends on another—or smoothing effects, which capture gradual, nonlinear trends over time. As observed in Figure 6, the parametric coefficients table summary, Mentor.CourseYes is one of the most significant factors in our predictive Z-score modeling. Meaning, mentored students that receive additional support are associated with a positive shift in their performance.

Also, all four subject indicators—Math, Science, English, and Business—have negative values compared to the baseline, which is the reference subject "Other." This means that, after accounting for mentoring, grade level, period, semester, absences, and the nonlinear effect of time, students tend to have slightly lower Z-Scores in these core subjects compared to electives and social-studies categorized as ‘Other’. This makes sense because students generally find electives easier. Because of the higher baseline difficulty, students in these four subjects generally start at a slightly lower Z‑score than they do in a typical elective or social‑studies class. Figure 7 illustrates these predicted Z-score trajectories for a typical student over time, highlighting the impact of mentoring, subject, and other variables included in the model. Parts of code from Visual Inspection of GAMM Models by van Rij (2016) was used to create these plots.

The main effect of grade year (+0.0442) shows that older students who are not receiving mentoring in a particular course but are still in the program tend to perform better than their younger peers in general, with each additional grade year associated with a slight increase in standardized performance. Yet, this overall trend shifts once we account for interaction terms, which reveal that the effect of mentoring changes depending on both the student’s grade year and the subject. As briefly mentioned earlier, interaction terms show how the combined influence of two factors differs from their individual effects. In this case, they reveal that the impact of mentoring is not the same for every grade level or subject; instead, mentoring may be more beneficial for some groups or subjects than others. For example, the interaction between mentoring and grade year is negative (–0.3034), suggesting that mentoring tends to have a stronger positive impact on younger students compared to older ones. Similarly, interactions reveal that mentoring effects differ by subject: mentoring in English shows a negative interaction (–0.702), while mentoring in Business has a positive effect (+0.3239). These differences indicate that mentoring effectiveness is influenced not only by the student’s grade level but also by the subject in which mentoring occurs.

Since some interaction effects in the model are complex and not easily interpreted from coefficients alone, visualizing predicted outcomes offers a clearer understanding. To do this, we generated predictions where some variables were held constant to represent a typical student, while others were allowed to vary in order to capture key effects of interest—such as mentoring, time, and subject. Below is a breakdown of how these variables were treated for prediction purposes:

Fixed Variables

These variables were held constant across all simulations to isolate the effect of mentoring, week, and mentoring frequency:

- Grade.Year (factor) – Set to the most frequent grade level (e.g., 10th grade) to represent a typical student.
- Period (factor) – Held at 1 to control for scheduling differences across class periods.
- Semester (factor) – Set to “S1” (Semester 1) to focus our scope within a single term.
- Total.Days.Absent (numeric) – Fixed at the mean absence value to account for typical attendance.

- Student.ID (NA / factor) – Set to NA or excluded to avoid student-specific random effects.

Variables We Allow to Vary

- s(Week, k=15) (numeric) – A smooth curve over weeks that can flexibly capture meaningful trends by allowing up to 15 bends without overreacting to random fluctuations
- Mentor.Course (Yes vs. No) × Course.Category(Factor)
- s(Running.Total..Class , k=6) (Numeric)

This allows us to predict what a student's performance *could* look like if mentoring is added or removed. It also gives us insight into rollover effects: whether mentoring in one course helps a student's progress in non-mentored courses. These broader impacts could potentially highlight mentoring if their effects provide a comprehensive influence to a student's academic performance. In short, these models help us focus our scope beyond basic questions such as "is mentoring good?" into more engaging questions like "how, when, and for who?" is mentoring making the most difference.

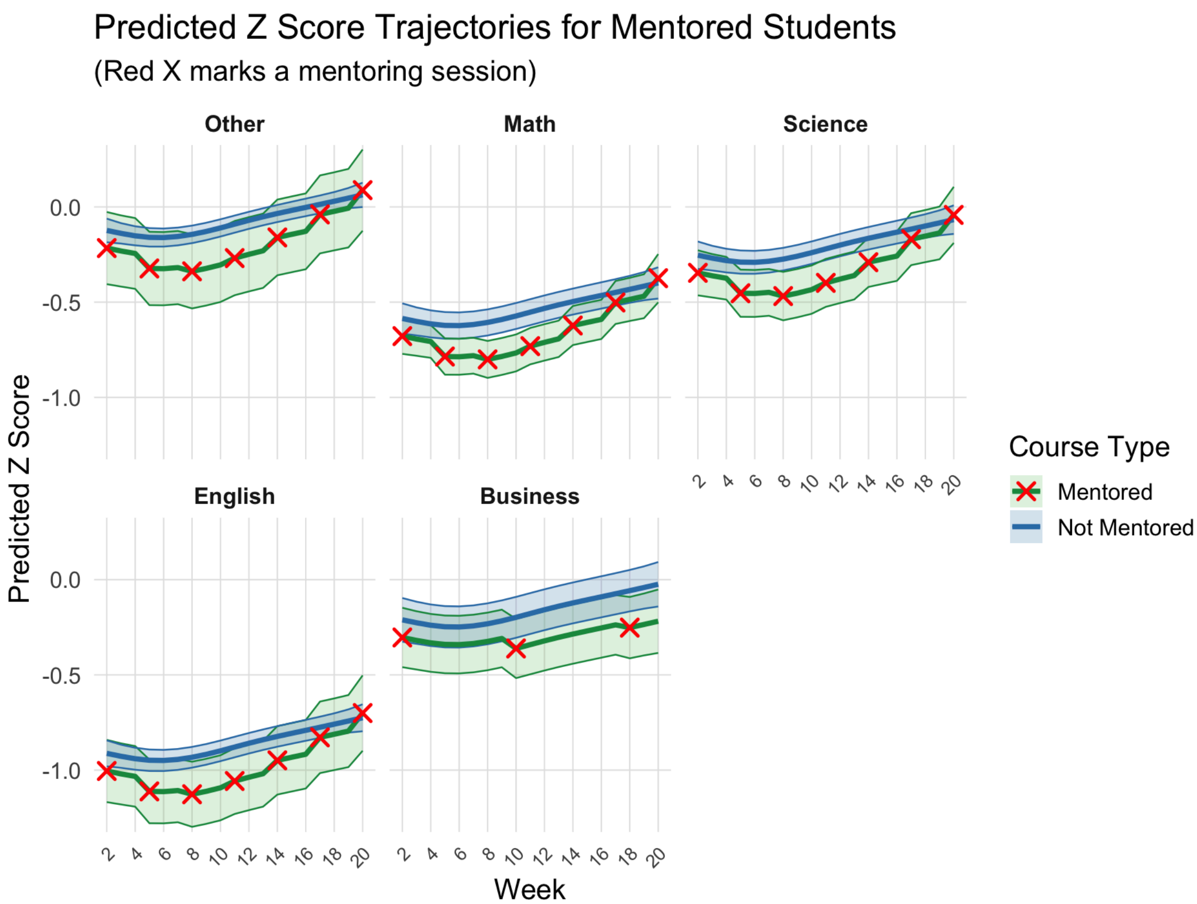

Figure 7

Predicted Grade Z Score Trajectories: Across All Subjects

Across all five subjects—Math, Science, English, Business, and Other—the objective is to evaluate both the direct effect of mentoring on student performance in that specific course and any potential spillover effects on other courses where students were not directly mentored. In Figures: 7-12, the curves represent the average performance of mentored students over time when receiving mentorship in a specific subject (green) and when not (blue). Students who receive course-specific mentoring tend to start with lower predicted Z-scores, which is expected since they are already struggling in those subjects. However, we observe that each mentoring session is followed by a noticeable "boost" or upward shift in their performance trajectory a few weeks later. For example when looking at the plot for Science in Figure 7, the week 8 mentoring session shows a lift by week 10, and then week 11 has that same boost upwards by week 13.

In Business classes, however, the typical gap between mentoring sessions was around 8 weeks. This is largely due to many students attending only a few sessions within a short timeframe, which lowered the average number of sessions per student. Since the model estimates gaps by distributing sessions evenly across the semester, this sparse attendance creates longer intervals between predicted boosts—resulting in a more plateaued trajectory until the next increase. For example, the inflection from week 2 mentoring extends for longer than the other subjects before we see another uptick. This demonstrates that over time, regular support from mentoring sessions accumulates and even accelerates a student's progress compared to students that aren't receiving help in that same course.

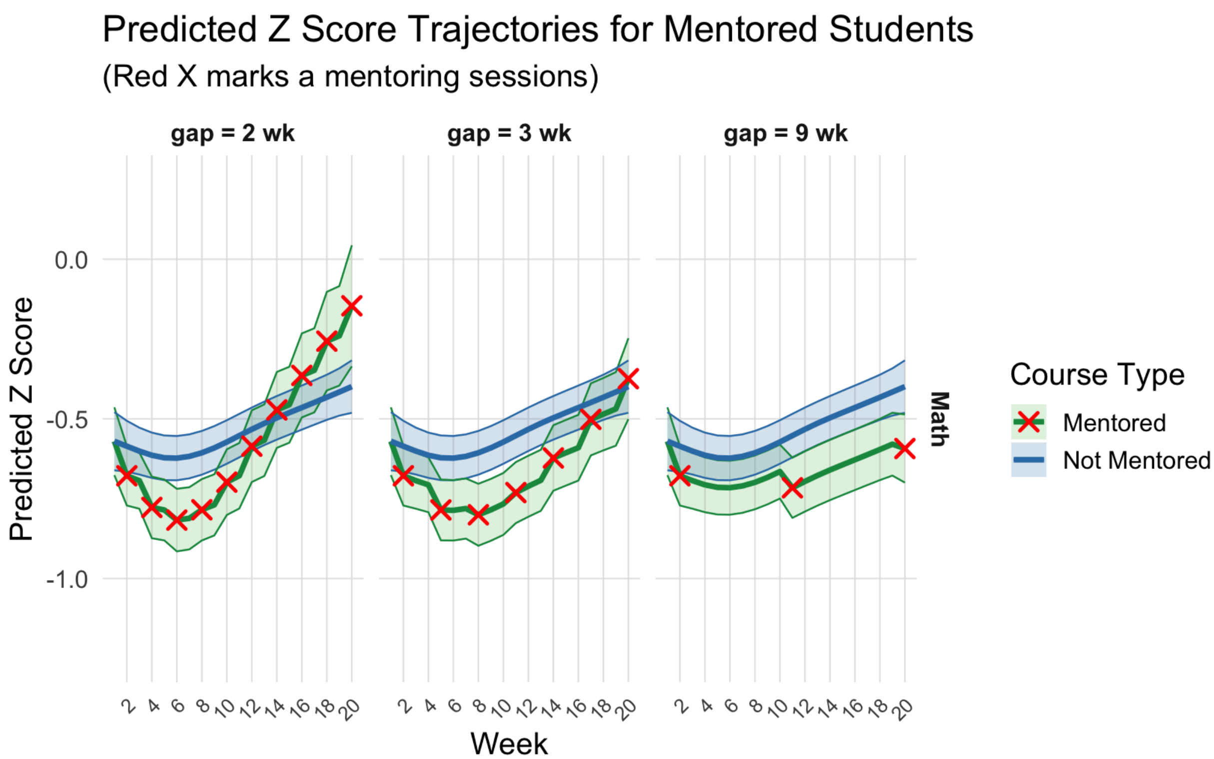

Figure 8

Predicted Grade Z Score Trajectories For "Best" and "Worst" Scenarios for Math

By using the same GAM we can now estimate each student's predicted behavior in a best, average, and worst case scenario. Figures: 8-12 show the trajectory of a typical student with different "dosages" of mentoring across all the subjects. In the case of Math, if a student goes to mentoring every 2 weeks, their predicted Z-score rises dramatically to the point of overtaking their peers and succeeding above the blue trend line. On the contrary, in a worst case scenario when sessions are more spaced out apart (students are less consistent with their mentoring) the improvements we see get delayed late into the term and move at a more gradual pace. Similar patterns can be observed in the figures for the other subjects below.

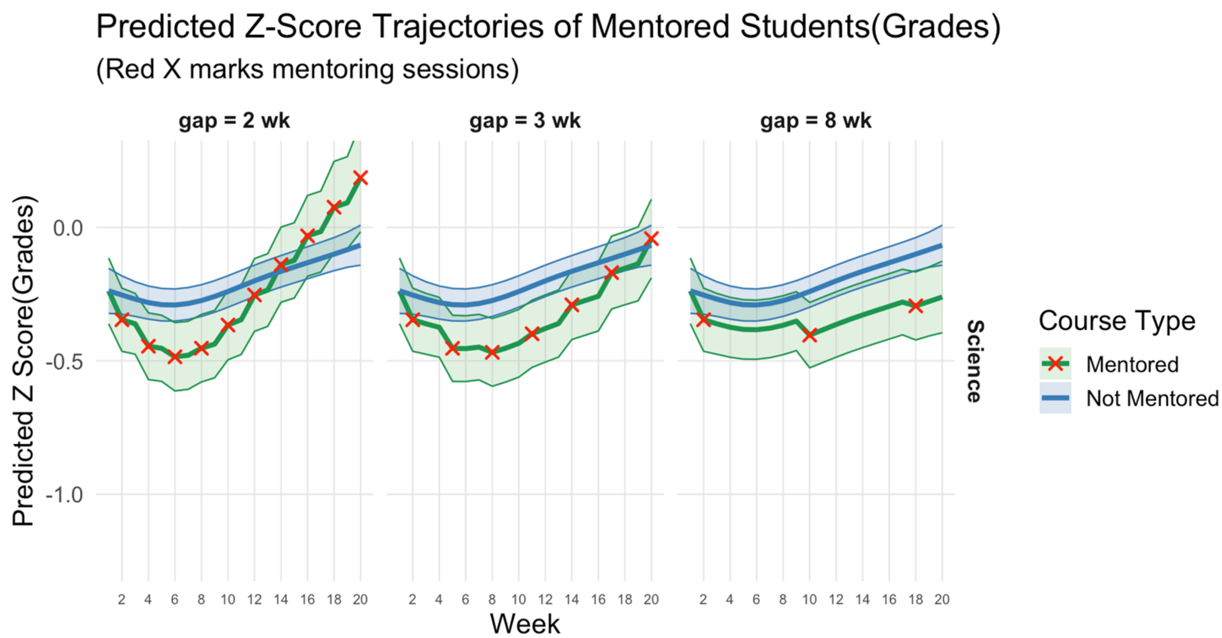

Figure 9

Predicted Grade Z Score Trajectories For "Best" and "Worst" Scenarios for Science

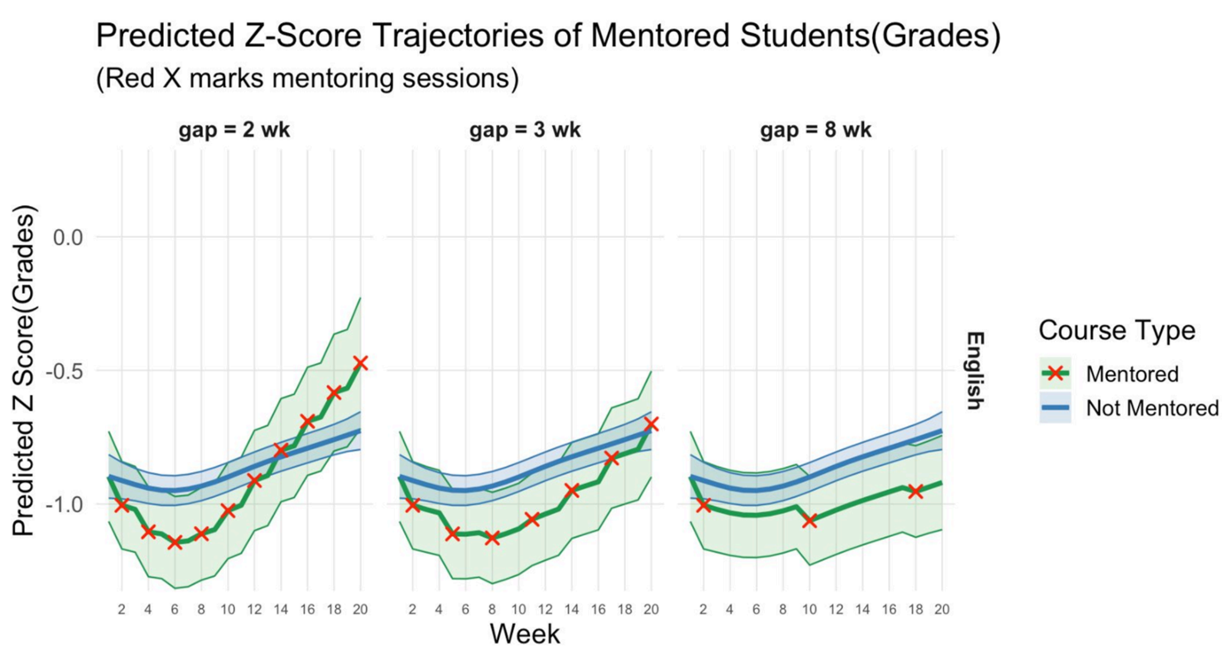

Figure 10

Predicted Grade Z Score Trajectories For "Best" and "Worst" Scenarios for English

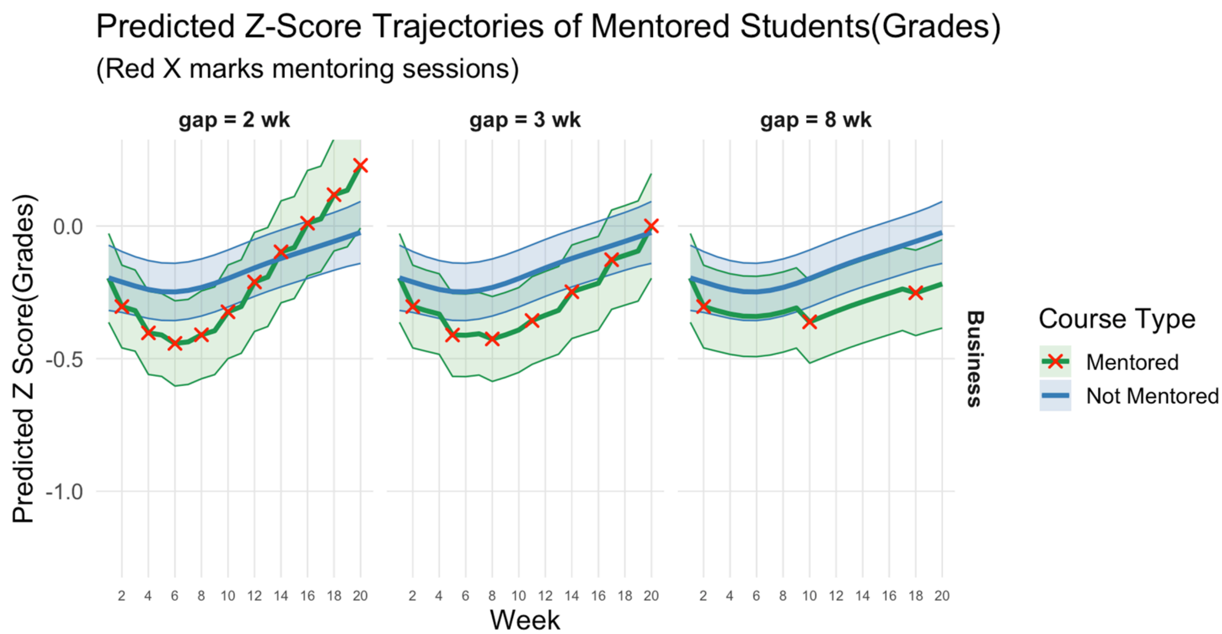

Figure 11

Predicted Grade Z Score Trajectories For "Best" and "Worst" Scenarios for Business

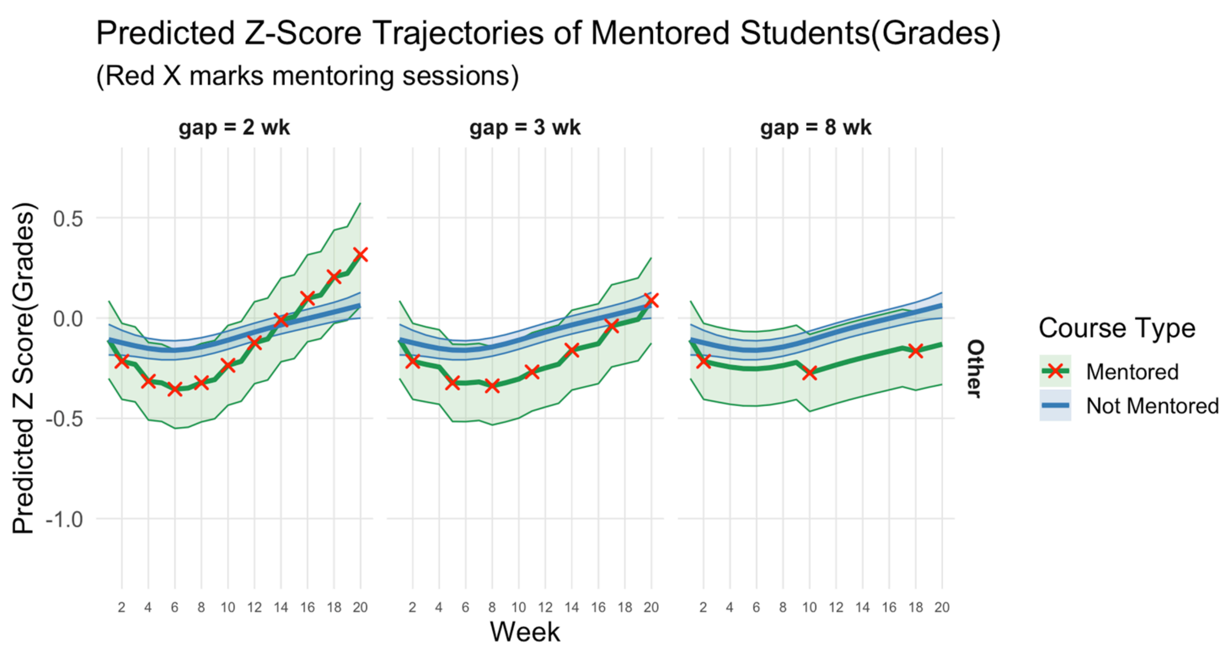

Figure 12

Predicted Grade Z Score Trajectories For "Best" and "Worst" Scenarios for Other

6 Conclusion and Next Steps

The effects of mentoring become especially clear and measurable when we can see consistent cumulative improvements over time. While each individual mentoring session may not lead to immediate short-term gains, the effects appear to build over time, with earlier and consistent sessions contributing to long-term improvement. We observe an upward trend in students' Z-scores at levels higher than where they started and often above those of their non-mentored peers.

Core subjects, however, tend to start with lower predicted Z-scores as they have negative coefficients. This can be attributed to how varied the teaching styles and assignments of electives can be taught. While these core subjects may take longer for students to show noticeable improvement and "catch up," model predictions indicate that students who attend mentoring sessions every two weeks are likely to outperform their non-mentored peers in most of them. Individual trajectories—such as the contrast between Student 6 and Student 32 in Geometry A and B—highlight how early mentoring support can have a sustained impact by building healthy habits and a support system for subsequent semesters.

The next step in the analysis is to extend the data review beyond Week 34 to evaluate end-of-year outcomes and gain a clearer picture of the program's long-term effects. One potential issue we identified is the presence of skewed data, particularly within the group of students who are mentored in other courses but not in the specific course being analyzed. In this group, grades and attendance tend to cluster at very high or very low values (as shown in Appendix C: Figures C1, C2, C3, and C4). This skewness could possibly affect the results, but we are not sure by how much. To address this, future analyses could consider applying percentile-based metrics to the control group—the students who are not mentored in any courses. These metrics could be used in place of Z-scores to interpret both grades and attendance more reliably. Because percentile ranks compare students to their peers within the same group, they provide a standardized view of performance and are less affected by outliers or skewed distributions. This makes them especially useful when evaluating control group data, where grades and attendance often cluster at extreme values. Although our model already performs reliably, incorporating these steps could further enhance its accuracy and offer a more nuanced understanding of mentoring's effects.

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Appendix A: Data Preparation and Cleaning Process

The data cleaning and preparation process was done in Python using the pandas library, which is widely used for organizing and analyzing structured tabular data.

The raw data consisted of five CSV files: two containing grade and attendance records for Semester 1 (Weeks 2–20), two for Semester 2 (Weeks 21–34), and one containing mentoring session logs. All the grade and attendance records were consistent in their formatting, but they did need to be joined. As for the mentoring session logs data, there was no joining necessary. However, the formatting did differ from the other datasets. Therefore, for consistent analysis across files, we began by aligning the formats of each dataset. This included standardizing column names (e.g., renaming variations like "Student #," "PSU#," and "Mentor Active" to "Student ID" and "Mentor Activity"), converting missing values (such as blanks or "na") to proper "NA" values, and removing stray characters or spacing.

The mentoring logs initially recorded each mentoring session as a separate column by day. To make this data analysis-ready, we reshaped it into long format, where each row represents one student, one course, and one week. Date strings (e.g., "Tues. 4/22") were parsed and aligned to that week's Monday for consistency. Attendance markers such as checkmarks were converted to categorical flags ("Yes," "No," or "Not Applicable"), and duplicate rows—typically caused by multiple sessions in a single week—were collapsed into a single record with an accurate count of sessions attended.

Similar things were done to the Grade records. "Grade Pulled" dates were normalized to Mondays, score strings such as "87 pts" were converted to numeric values, and missing letter grades were filled in using a GPA-to-letter-grade mapping. The cleaned grade and attendance records were then merged with the mentoring data, so that each row held the full academic and mentoring information for a given student, course, and week.

For consistency, we used the structure of the grade and attendance dataset as the formatting standard and integrated important mentoring details into it. This allowed us to do a clear, week-by-week evaluation of mentoring participation and academic outcomes.

To support our analysis, we generated several derived indicators, including:

- Mentor Course: Whether a student was ever mentored in that specific course
- Mentored Student: Whether the student ever attended a mentoring session

- Running Totals: Cumulative mentoring sessions attended per class or semester or throughout the academic year

- Recency Counters: Weeks since a student's last session (overall and by course)

We also calculated z-scores for grades and absences to help identify how each student's performance compared to their classmates in the same course and week in these areas. Finally, we performed quality checks, such as removing duplicate records from known data-entry errors (e.g., entire class lists accidentally copied twice), and filling in any missing weeks with placeholder rows to ensure each student-course combination had a complete timeline across the academic year.

Appendix B: Data Exploration

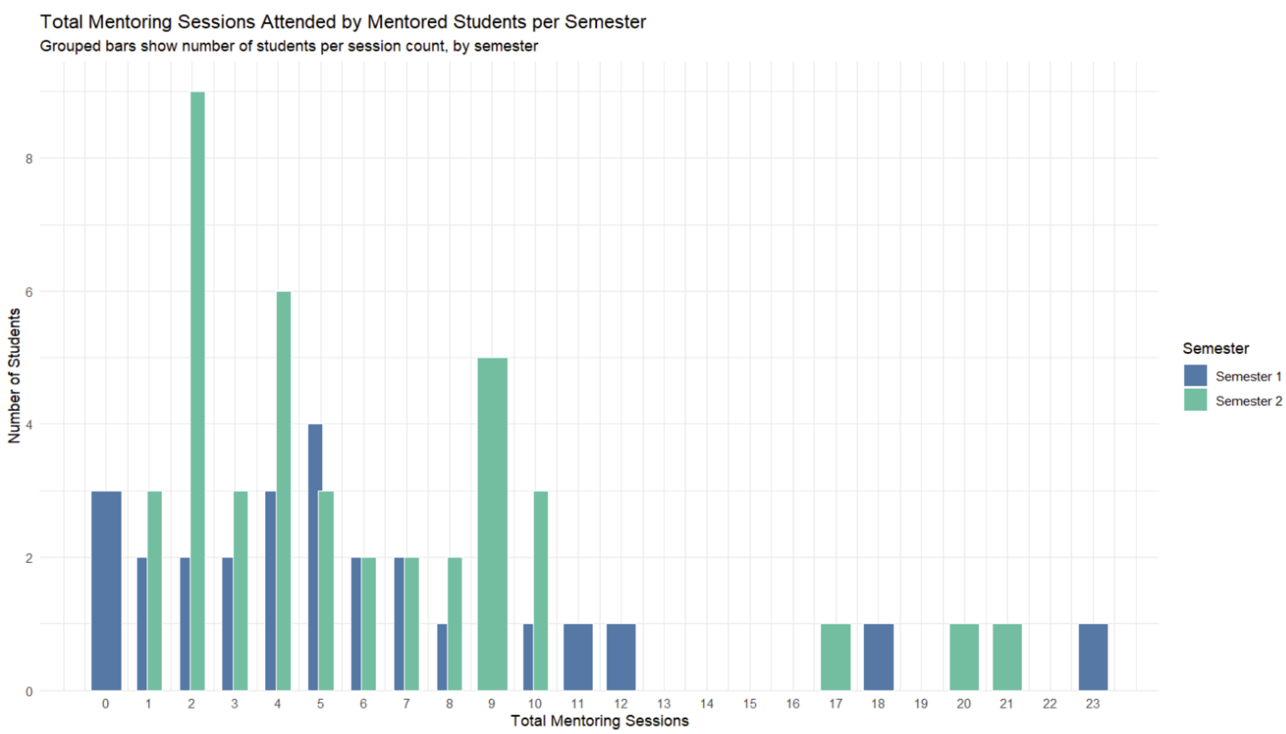

Figure B1

Total Mentoring Sessions Attended by Mentored Students per Semester

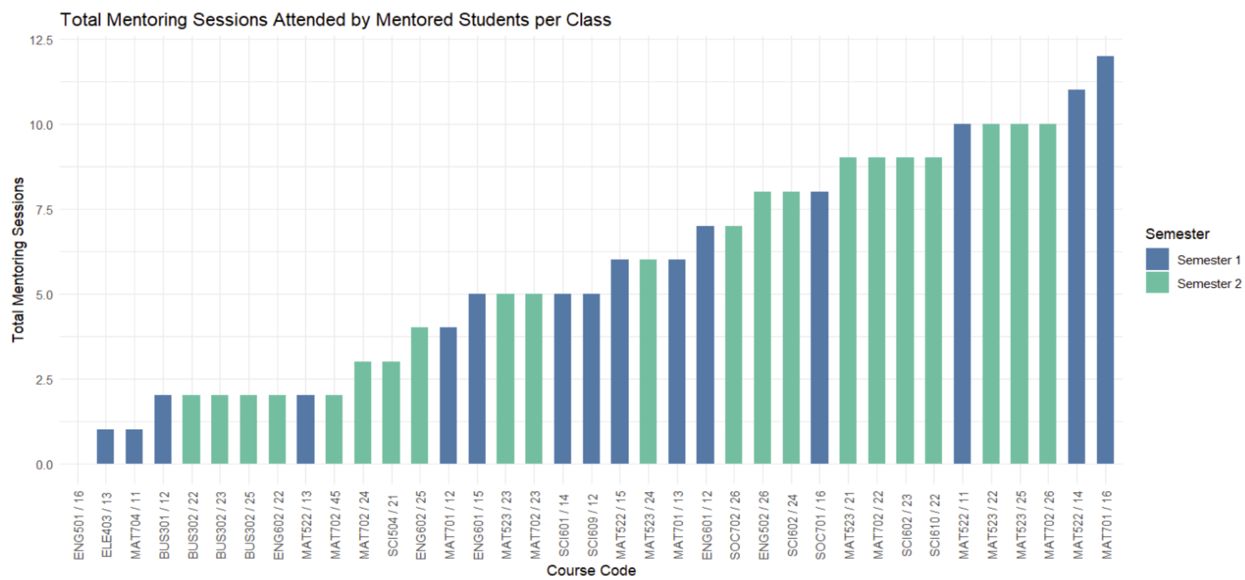

Figure B2

Total Mentoring Sessions Attended by Mentored Students per Class

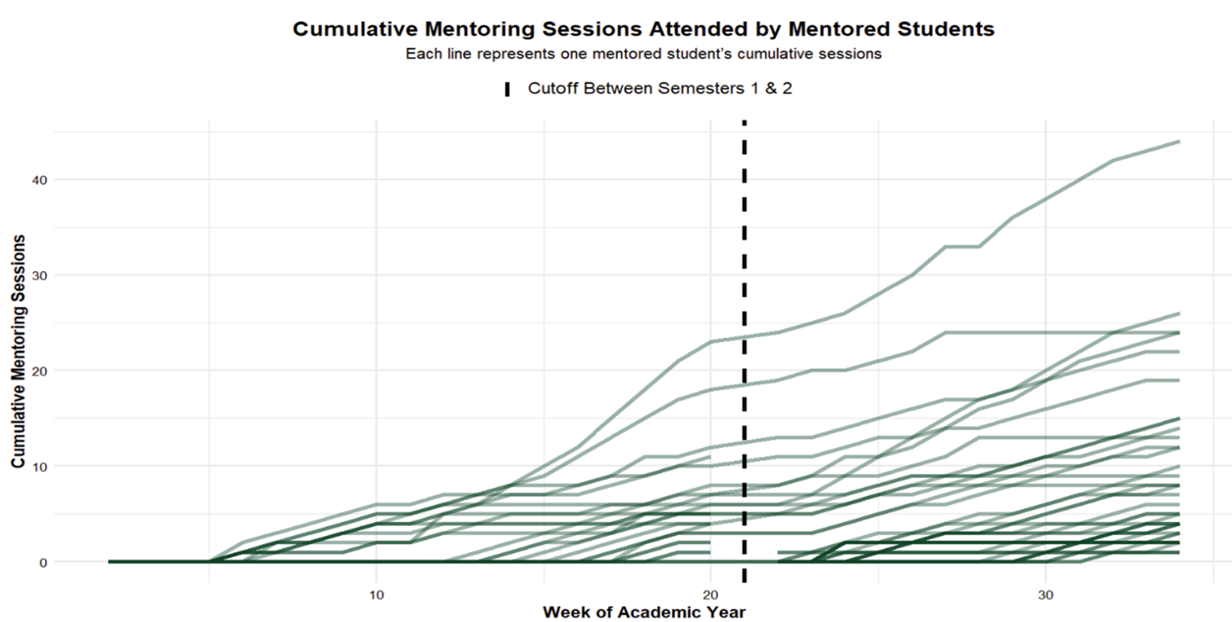

Figure B3

Cumulative Mentoring Sessions Attended by Mentored Students

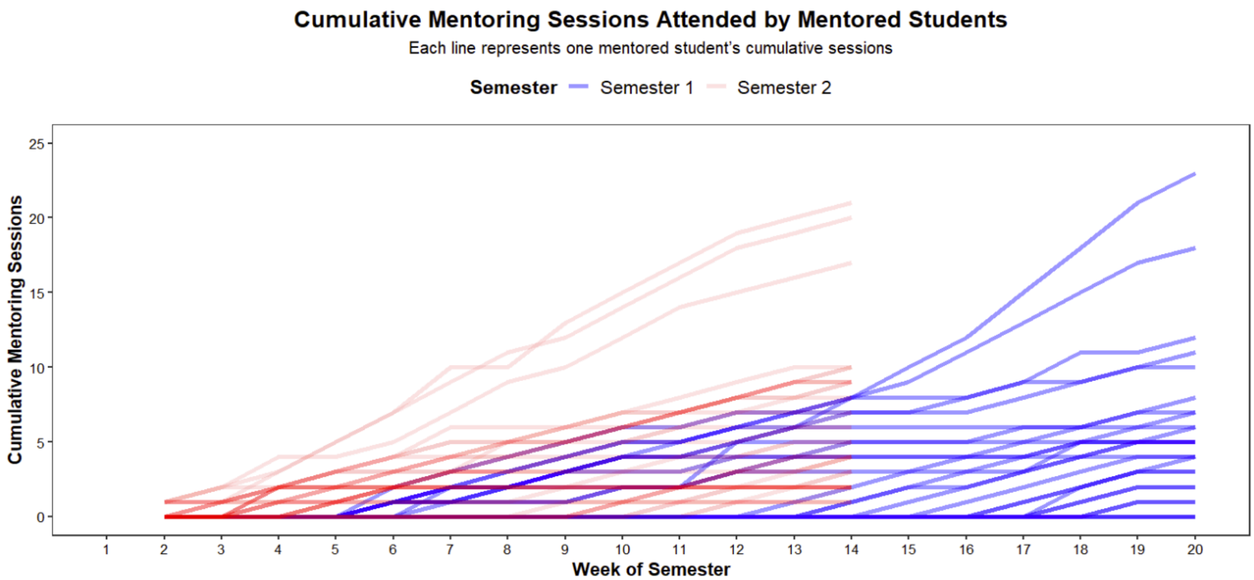

Figure B4

Cumulative Mentoring Sessions Attended by Mentored Students

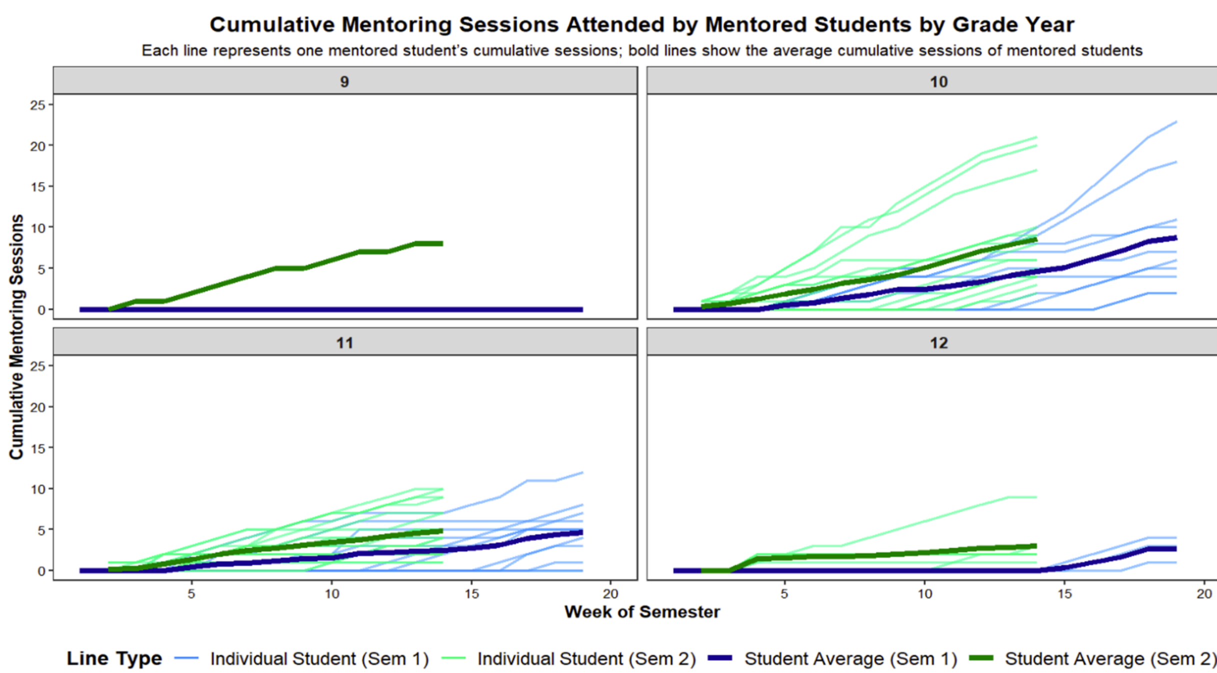

Figure B5

Cumulative Mentoring Sessions Attended by Mentored Students by Grade Year

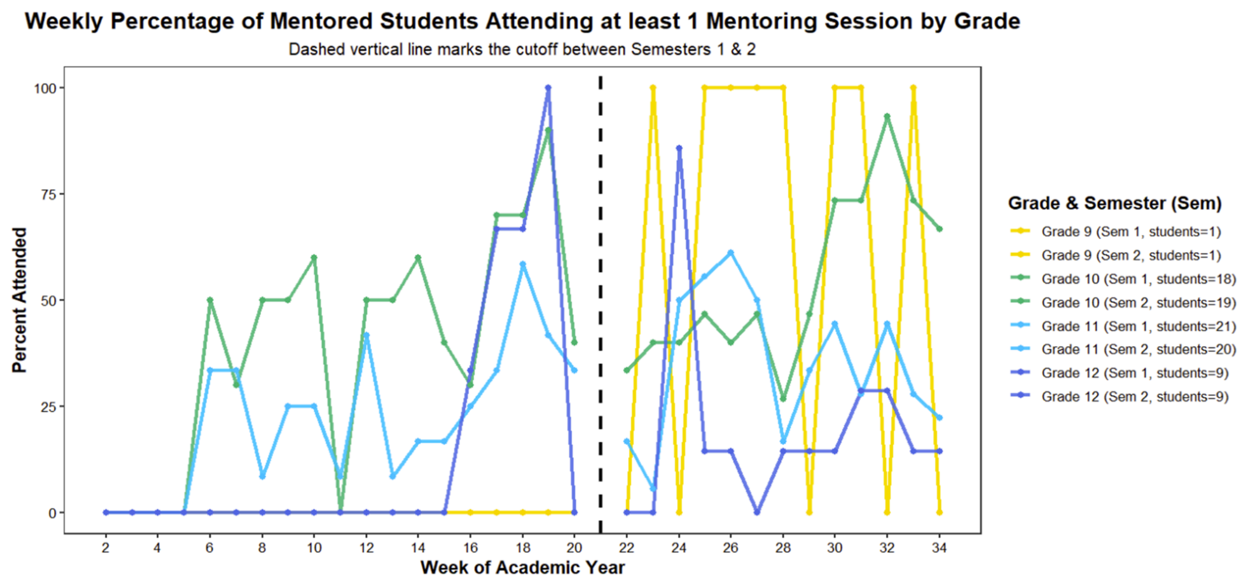

Figure B6

Weekly Percentage of Mentored Students Attending at least 1 Mentoring Session by Grade

Appendix C: Conclusion and Next Steps

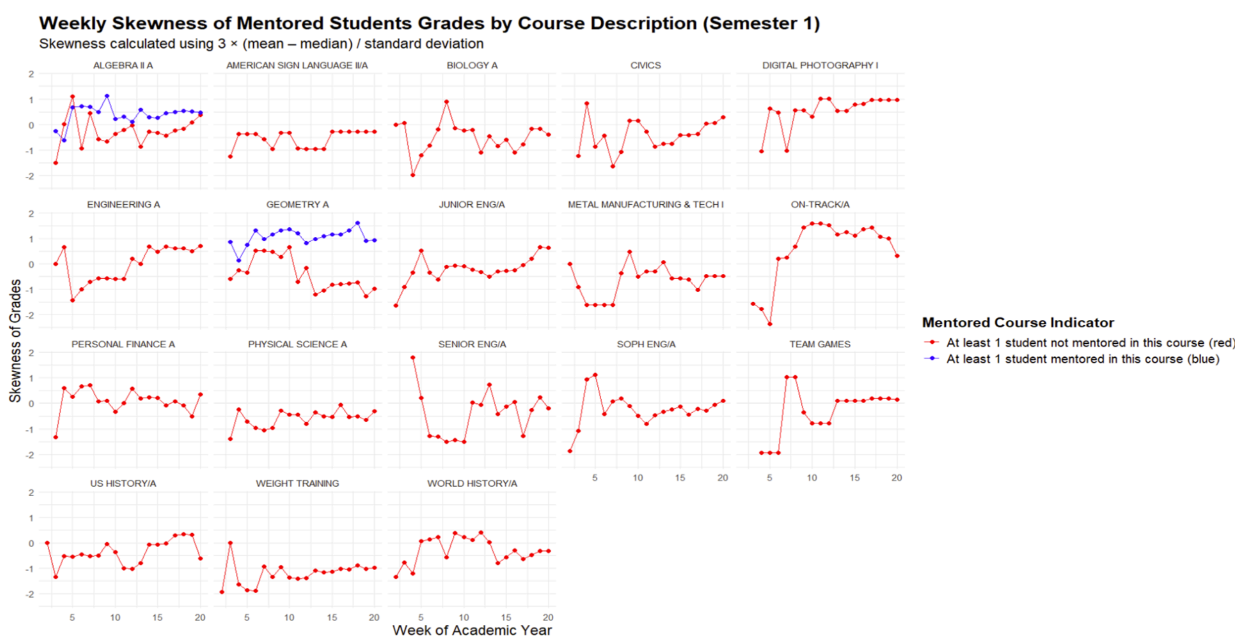

Figure C1

Weekly Skewness of Mentored Students Grades by Course Description (Semester 1)

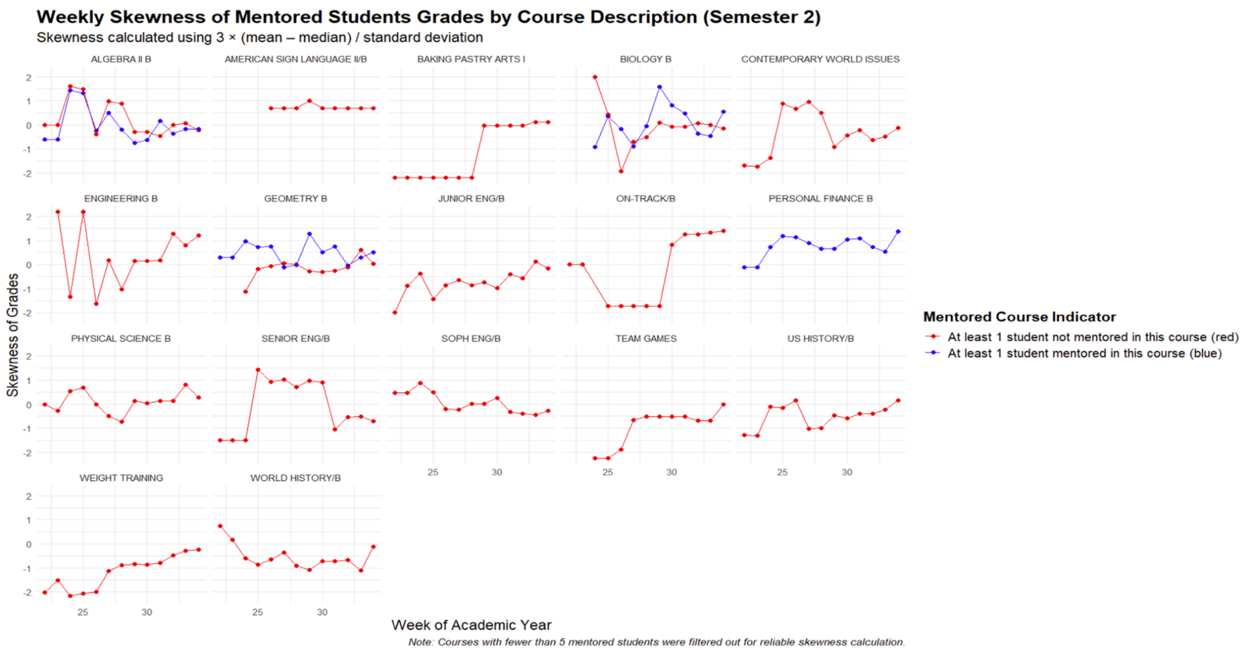

Figure C2

Weekly Skewness of Mentored Students Grades by Course Description (Semester 2)

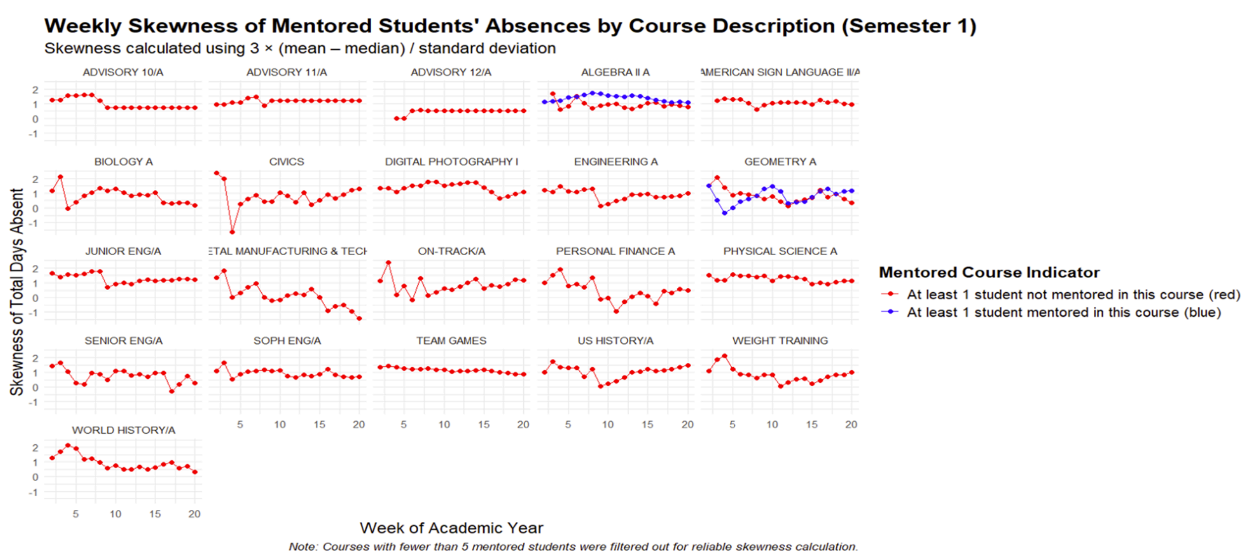

Figure C3

Weekly Skewness of Mentored Students' Absences by Course Description (Semester 1)

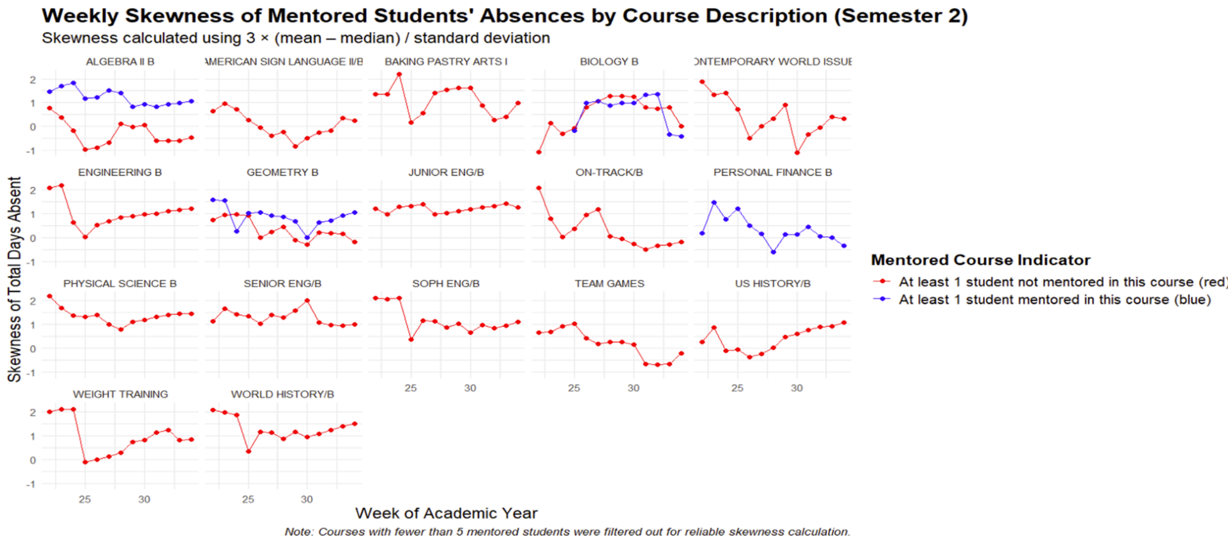

Figure C4

Weekly Skewness of Mentored Students' Absences by Course Description (Semester 2)